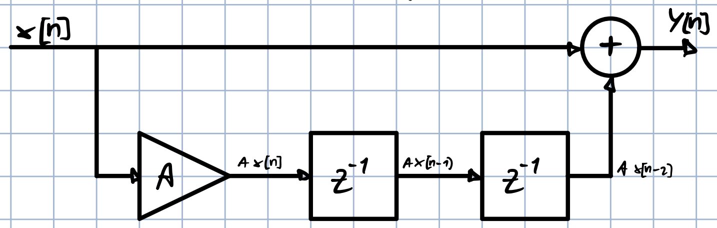
RECURSIVE SYSTEMS

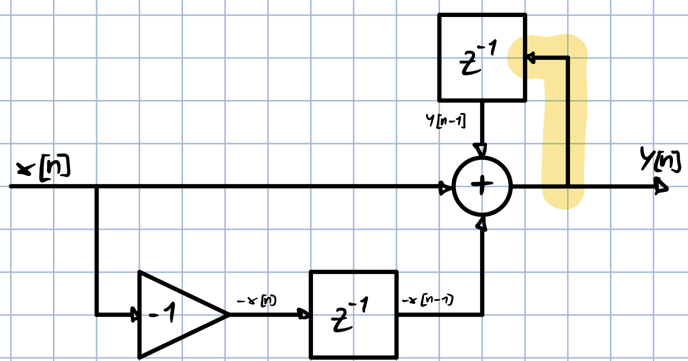
A SYSTEM IS DEFINED NON-RECURSIVE WHEN THE OUTPUT DEPENDS ONLY ON INPUTS, AND IN PARTICULAR PAST AND PRESENT ONES.

$$y[n] = x[n] + A\,x[n-2]$$



A SYSTEM IS DEFINED RECURSIVE WHEN THE OUTPUT DEPENDS NOT ONLY ON INPUTS, BUT ALSO ON PAST OUTPUT VALUES

$$y[n] = x[n] - x[n-1] + y[n-1]$$



LET'S SUPPOSE TO WORK WITH AN LTI FIR SYSTEM. IN GENERAL FOR AN LTI SYSTEM THE CONVOLUTION EXISTS:

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]\,h[n-k] = \sum_{k=-\infty}^{+\infty} h[k]\,x[n-k]$$

SINCE THE SYSTEM IS CAUSAL $(h[n]=0 \;\; \forall n<0)$ AND FIR

$$y[n] = \sum_{\underset{\text{causal}}{k=0}}^{+\infty} h[k]\,x[n-k] = \sum_{k=0}^{\overset{\text{FIR}}{M-1}} h[k]\,x[n-k]$$

FOR SIMPLICITY LET'S SUPPOSE TO HAVE $h[n]$ OF LENGTH $M+1$

$$y[n] = \sum_{k=0}^{M} h[k]\,x[n-k]$$

$h[n]$ CAN BE SEEN AS A SIMPLE SET OF COEFFICIENTS b_k

$$y[n] = \sum_{k=0}^{M} b_k[k]\, x[n-k]$$

REPRESENTATION OF A NON-RECURSIVE SYSTEM, SINCE THE OUTPUT DEPENDS ONLY ON THE INPUTS

IF INSTEAD NOW WE WANT A GENERAL DESCRIPTION OF A RECURSIVE SYSTEM

$$\boxed{\sum_{k=0}^{N} a_k[k]\, y[n-k] = \sum_{k=0}^{M} b_k[k]\, x[n-k]}$$

An Nth-order linear constant-coefficient difference equation (LCCDE) is of the form

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] - \sum_{k=1}^{N} a_k y[n-k] \qquad (1)$$

It is *linear* because the sequences $x[n]$ and $y[n]$ appear linearly in (1), and it has constant coefficients because the coefficients a_k and b_k do not depend on the index n.

LCCDEs are important because they can be used to describe many practically useful discrete-time (or discrete-space) systems, such as linear time-invariant (LTI) filters (which are also called *linear shift-invariant* (LSI) filters, if n does not represent a time index).

Given appropriate initial conditions, Eq. (1) can be used to recursively compute the sequence $y[n]$ given its past values and the current and past values of the sequence $x[n]$. For causal LTI filters, $y[n]$ is interpreted as the filtered output, and $x[n]$ is the input. In that case, Eq. (1) describes an infinite impulse response (IIR) filter.

If $a_k = 0, k = 1, \dots, N$, Eq. (1) reduces to

$$y[n] = \sum_{k=0}^{M} b_k x[n-k] \qquad (2)$$

which is a convolution sum, and the coefficients b_k represent the system's impulse response. Note that the impulse response has finite length $M+1$, i.e., the system described by (2) is a finite impulse response (FIR) filter.

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WHICH CAN BE RE-WRITTEN AS

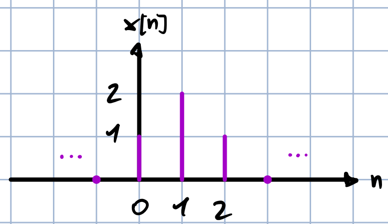
$$a_0 y[n] + \underbrace{\sum_{k=1}^{N} a_k[k]\, y[n-k]}_{\text{IF } =0 \text{ IT IS NON-RECURSIVE}} = \sum_{k=0}^{M} b_k[k]\, x[n-k]$$

$$\underset{\text{PRESENT OUTPUT}}{y[n]} = \frac{\overbrace{\sum_{k=0}^{M} b_k[k]\, x[n-k]}^{\text{INPUT}} - \overbrace{\sum_{k=1}^{N} a_k[k]\, y[n-k]}^{\text{PAST OUTPUTS}}}{a_0}$$

EX

$$y[n] = y[n-1] + x[n]$$

$y[-1] = \emptyset$ INITIAL CONDITION



$y[0] = y[-1] + x[0] = 0 + 1 = 1$

$y[1] = y[0] + x[1] = 1 + 2 = 3$

$y[2] = y[1] + x[2] = 3 + 1 = 4$

$y[3] = y[2] + x[3] = 4 + 0 = 4$

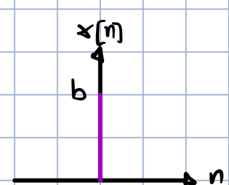
$y[n] = 4 \;\; \forall n \geq 4$

$y[-1] = y[-2] + x[-1] \rightarrow 0 = y[-2] + 0$

EX

$$y[n] = \alpha\, y[n-1] + x[n]$$

$$x[n] = b\,\delta[n]$$



$y[-1] = 1$

$$\left.\begin{aligned} y[0] &= \alpha\, y[-1] + x[0] = \alpha + b \\ y[1] &= \alpha\, y[0] + x[1] = \alpha(\alpha + b) + 0 \\ y[2] &= \alpha^2(\alpha + b) \end{aligned}\right\} \; y[n] = \alpha^n(\alpha + b) = \alpha^{n+1} + \alpha^n b \qquad \forall n \geq 0$$

$$\left.\begin{array}{l} y[-1] = \alpha\, y[-2] + x[-1] \rightarrow 1 = \alpha\, y[-2] + 0 \rightarrow y[-2] = \frac{1}{\alpha} \\ y[-2] = \alpha\, y[-3] + x[-2] \rightarrow y[-3] = \frac{1}{\alpha^2} \\ y[-3] = \alpha\, y[-4] + x[-3] \rightarrow y[-4] = \frac{1}{\alpha^3} \end{array}\right\} \; y[n] = \frac{1}{\alpha^{-n-1}} = \alpha^{n+1} \quad \forall n < 0$$

CHECK LTI:

$$x_1[n] = b\,\delta[n-1]$$

$$\left.\begin{array}{l} y[0] = \alpha\, y[-1] + x_1[0] = \alpha + 0 \\ y[1] = \alpha\, y[0] + x_1[1] = \alpha^2 + b \\ y[2] = \alpha\, y[1] + x_1[2] = \alpha(\alpha^2 + b) \\ y[3] = \alpha\, y[2] + x_1[3] = \alpha^2(\alpha^2 + b) \end{array}\right\} \; y_1[n] = \alpha^{n-1}(\alpha^2 + b) \quad \forall n \geq 1$$

$$\hookrightarrow \underbrace{\alpha^{n+1} u[n]}_{\text{FROM } n=0} + \underbrace{\alpha^{n-1} b\, u[n-1]}_{\text{FROM } n=1} \quad \forall n \geq 0$$

n	$y[n]$	$y_1[n]$
0	$\alpha + b$	α
1	$\alpha^2 + \alpha b$	$\alpha^2 + b$
2	$\alpha^3 + \alpha^2 b$	$\alpha^3 + \alpha b$
3	$\alpha^4 + \alpha^3 b$	$\alpha^4 + \alpha^2 b$
4	$\alpha^5 + \alpha^4 b$	$\alpha^5 + \alpha^3 b$

$y_1[n] \neq y[n-1] \implies$ NOT LTI

LET'S NOW SUPPOSE TO CHANGE THE INITIAL CONDITIONS:

$$y[n] = \alpha\, y[n-1] + x[n]$$

$$x[n] = b\,\delta[n]$$

$$y[-1] = 0$$

$$\left.\begin{array}{l} y[0] = b \\ y[1] = \alpha b \\ y[2] = \alpha^2 b \end{array}\right\} \; y[n] = \alpha^n b \quad \forall n \geq 0$$

$$\left.\begin{array}{l} y[-1] = \alpha\, y[-2] \rightarrow y[-2] = \frac{y[-1]}{\alpha} = 0 \\ y[-2] = \alpha\, y[-3] \rightarrow y[-3] = \frac{y[-2]}{\alpha} = 0 \end{array}\right\} \; y[n] = 0 \quad \forall n < 0$$

CHECK TI:

$$x_1[n] = x[n-1] = b\,\delta[n-1]$$

$$\left.\begin{array}{l} y_1[0] = \alpha\, y[-1] + b\,\delta[-1] = 0 \\ y_1[1] = \alpha\, y[0] + b = b \\ y_1[2] = \alpha\, y[1] = \alpha b \\ y_1[3] = \alpha^2 b \end{array}\right\} \; y_1[n] = \alpha^{n-1} b\, u[n-1] \quad \forall n \geq 0$$

$$\left.\begin{array}{l} y[-1] = \alpha\, y[-2] \rightarrow y[-2] = \frac{y[-1]}{\alpha} = 0 \\ y[-2] = \alpha[y-3] \rightarrow y[-3] = \frac{y[-2]}{\alpha} = 0 \end{array}\right\} \; y_1[n] = 0 \quad \forall n < 0$$

n	$y[n]$	$y_1[n]$
0	b	0
1	ab	b
2	$a^2 b$	ab
3	$a^3 b$	$a^2 b$
4	$a^4 b$	$a^3 b$

$y_1[n] = y[n-1] \Rightarrow$ IT IS TIME-INVARIANT

! THE INITIAL CONDITIONS CAN SIGNIFICANTLY INFLUENCE THE SYSTEM

LCCDEs

LINEAR CONSTANT COEFFICIENTS DIFFERENCE EQUATIONS CAN BE USED TO REPRESENT LTI SYSTEMS.
FOR CAUSAL LTI SYSTEMS, INITIAL CONDITIONS AND INPUT ARE CONSIDERED TO BE $0 \; \forall n < 0$

GIVEN AN LCCDE AS THE I/O RELATIONSHIP DESCRIBING A CAUSAL LTI SYSTEM, THE GOAL IS TO DERIVE AN EXPLICIT EXPRESSION FOR $y[n]$, $n \geq 0$ WITH RESPECT TO $x[n]$ AND A SET OF INITIAL CONDITIONS AND IT'S IMPULSE RESPONSE.

ONE METHOD IS TO ASSUME THE SOLUTION AS THE SUM OF TWO SUB-SOLUTIONS: HOMOGENEOUS AND PARTICULAR.

$$y[n] = y_h[n] + y_p[n]$$

HOMOGENEOUS SOLUTION

NOT $x[n] = 0$

THE HOMOGENEOUS SOLUTION IS THE SOLUTION WITH ALL INPUT COEFFICIENTS ZERO $(b_k = 0 \; \forall k)$.
IT ISOLATES THE SYSTEM'S INTRINSIC "RESPONSE".

↳ JUST EFFECT OF PAST OUTPUTS, WHICH ARE INTRINSIC TO THE SYSTEM AND DON'T DEPEND ON INPUT

$$\sum_{k=0}^{N} a_k y[n-k] = 0 \quad \longleftrightarrow \quad a_0 y[n] + a_1 y[n-1] + \cdots + a_N y[n-N] = 0$$

WE CAN WRITE IT AS A POLYNOMIAL ASSUMING THE OUTPUT AS AN EXPONENTIAL

$$y[n] := \lambda^n$$

$$\downarrow$$

CHARACTERISTIC POLYNOMIAL

$$\sum_{k=0}^{N} a_k \lambda^{n-k} = \sum_{k=0}^{N} a_k \lambda^{N-k} \lambda^{n-N} = \lambda^{n-N} \sum_{k=0}^{N} a_k \lambda^{N-k} \longrightarrow \boxed{\sum_{k=0}^{N} a_k \lambda^{N-k} = 0} = a_0 \lambda^N + a_1 \lambda^{N-1} + a_2 \lambda^{N-2} + \cdots$$

$n-k = n-k+N-N = (N-k)+(n-N)$

THE CHARACTERISTIC POLYNOMIAL HAS N ROOTS, WHICH ARE THE CHARACTERISTIC/NATURAL MODES OF THE LTI SYSTEM
THEY CAN BE REAL OR COMPLEX (IN CONJUGATE PAIRS) AND MULTIPLE ROOTS MAY BE EQUAL (OVERLAPPING).

IF THE ROOTS ARE DISTINCT:

$$y_{hom}[n] = C_1 \lambda_1^n + C_2 \lambda_2^n + \cdots + C_N \lambda_N^n \quad , \; n \geq 0$$

WHERE $C_1, \dots, C_N$ ARE WEIGHTING COEFFICIENTS DETERMINED BY INITIAL CONDITIONS AFTER COMPUTING THE PARTICULAR SOLUTION AND λ_i ARE THE SOLUTIONS OF THE POLYNOMIAL

IF INSTEAD A ROOT (E.G. λ_1) HAS MULTIPLICITY M:

$$Y_{hom}[n] = C_1 \lambda_1^n + C_2 n \lambda_1^n + C_3 n^2 \lambda_1^n + \dots + C_M n^{M-1} \lambda_1^n + C_{M+1} \lambda_2^n + \dots + C_N \lambda_{N-M}^n, \; n \geq 0$$

PARTICULAR SOLUTION

$$\sum_{k=0}^{N} a_k[k]\, Y_p[n-k] = \sum_{k=0}^{M} b_k[k]\, x[n-k]$$

TO SOLVE IT WE ASSUME THAT $Y_p[n]$ DEPENDS ON THE FORM OF $x[n]$.
IF $x[n]$ IS AN EXPONENTIAL, WE ASSUME THE PARTICULAR SOLUTION TO BE AN EXPONENTIAL TOO.

$$x[n] = \alpha\, u[n] \implies Y_p[n] = \beta\, u[n]$$

$$x[n] = \alpha \gamma^n u[n] \implies Y_p[n] = \beta \gamma^n u[n]$$

EX

$$Y[n] + Y[n-1] - 6Y[n-2] = x[n]$$

$Y[-1] = 1$
$Y[-2] = -1$

1) HOMOGENEOUS

$a_0 = 1$
$a_1 = 1$
$a_2 = -6$
$b_0 = 1$

$$\sum_{k=0}^{N} a_k[k]\, Y[n-k] = \cancel{\sum_{k=0}^{M} b_k[k]\, x[n-k]}^{=0} \longrightarrow Y[n] + Y[n-1] - 6Y[n-2] = 0$$

2) POLYNOMIAL

$$\sum_{k=0}^{N} a_k \lambda^{N-k} = 0 \xrightarrow{N=2} a_0\lambda^2 + a_1\lambda + a_2 = 0 \longrightarrow \lambda^2 + \lambda - 6 = 0 \longrightarrow \lambda_{1,2} = \frac{-1 \pm \sqrt{1+24}}{2} \begin{cases} 2 \\ -3 \end{cases}$$

$$Y_{hom}[n] = C_1 \cdot (-3)^n + C_2 \cdot (2)^n$$

3) PARTICULAR

$x[n] = 8\, u[n]$
$Y_p[n] = \beta\, u[n]$

$$Y_p[n] + Y_p[n-1] - 6Y_p[n-2] = x[n]$$

$$y_p[n] + y_p[n-1] - 6\,y_p[n-2] = 8\,\mu[n] \quad \Big\downarrow\, x[n]=8\mu[n]$$

$$\Big\downarrow\, y_p=\beta\mu[n]$$

$$\beta\,\mu[n] + \beta\,\mu[n-1] - 6\,\beta\,\mu[n-2] = 8\,\mu[n]$$

$$\Big\downarrow\, @n=2$$

$$\beta + \beta - 6\beta = 8$$

$$-4\beta = 8$$

$$\beta = -2 \longrightarrow y_p[n] = -2\,\mu[n]$$

4) Full solution

$$y[n] = y_h[n] + y_p[n] = C_1\cdot(-3)^n + C_2\cdot(2)^n - 2\,\mu[n]$$

$y[-1] = 1$
$y[-2] = -1$

$$\begin{cases} C_1\,(-3)^{-1} + C_2\,(2)^{-1} - 2\,\mu[-1] = 1 \\ C_1\,(-3)^{-2} + C_2\,(2)^{-2} - 2\,\mu[-2] = -1 \end{cases} \quad \begin{cases} -\frac{C_1}{3} + \frac{C_2}{2} - 1 = 0 \\ \frac{C_1}{9} + \frac{C_2}{4} + 1 = 0 \end{cases} \quad \begin{cases} C_1 = \frac{3}{2}C_2 - 3 \\ C_1 = -\frac{9}{4}C_2 - 9 \end{cases}$$

$$\frac{3}{2}C_2 - 3 = -\frac{9}{4}C_2 - 9$$

$$\frac{3}{2}C_2 + \frac{9}{4}C_2 = -9 + 3$$

$$\frac{15}{4}C_2 = -6$$

$$C_2 = \frac{-24}{15} = -\frac{8}{5}$$

$$C_1 = -\frac{3}{2}\frac{8}{5} - 3 = -\frac{27}{5}$$

$$C_1 = \frac{9}{4}\frac{8}{5} - 9 = -\frac{27}{5}$$

THIS IS THE WRONG WAY TO DO IT!

Because $y[n]$ is valid only for $n \geq 0$??

$$y[n] = -\frac{27}{5}(-3)^n - \frac{8}{5}(2)^n - 2\,\mu[n]$$

$$y[n] + y[n-1] - 6\,y[n-2] = x[n]$$

$$\begin{cases} y[0] + y[-1] - 6\,y[-2] = x[0] \\ y[1] + y[0] - 6\,y[-1] = x[1] \end{cases} \xrightarrow{\text{plug in initial conditions}} \begin{cases} y[0] + 1 + 6 = x[0] \\ y[1] + y[0] - 6 = x[1] \end{cases} \xrightarrow{x[n]=8\mu[n]} \begin{cases} y[0] + 1 + 6 = 8\,\mu[0] = 8 \\ y[1] + y[0] - 6 = 8\,\mu[1] = 8 \end{cases}$$

Now find $y[0]$ and $y[1]$

$$y[0] = C_1\,(-3)^0 + C_2\,(2)^0 - 2\,\mu[0] = C_1 + C_2 - 2$$

$$y[1] = C_1\,(-3)^1 + C_2\,(2)^1 - 2\,\mu[1] = -3C_1 + 2C_2 - 2$$

$$\begin{cases}(C_1+C_2-2)+1+6=8\\(-3C_1+2C_2-2)+(C_1+C_2-2)-6=8\end{cases}\quad\begin{cases}C_1+C_2=3\\-2C_1+3C_2=18\end{cases}\quad\begin{cases}C_1=3-C_2\\-6+2C_2+3C_2=18\end{cases}\quad\begin{cases}C_1=-\frac{9}{5}\\C_2=\frac{24}{5}\end{cases}$$

$$y[n]=-\frac{9}{5}(-3)^n+\frac{24}{5}(2)^n-2\,u[n]\quad,\quad n\geq 0$$

$$y[n]=-\frac{9}{5}(-3)^n+\frac{24}{5}(2)^n-2\quad,\quad n\geq 0$$

ZERO INPUT AND ZERO STATE

THE TOTAL SOLUTION CAN ALSO BE SEEN AS THE SUM OF THE ZERO INPUT AND THE ZERO STATE SOLUTIONS.

ZERO INPUT: THE HOMOGENEOUS SOLUTION WITH COEFFICIENTS C_i OBTAINED FROM THE INITIAL CONDITIONS WITHOUT ADDING THE PARTICULAR SOLUTION

ZERO STATE: THE SOLUTION WITH INITIAL CONDITIONS $=0\ \forall n<0$

$$y[n]=y_{zi}[n]+y_{zs}[n]$$

EX

SOLVE WITH THE TWO METHODS

$y[n]-3y[n-1]=x[n]$

$y[-1]=1$

$x[n]=2u[n]$

1.1) HOMOGENEOUS

$$y[n]-3y[n-1]=0\longrightarrow\lambda-3=0\rightarrow\lambda=3$$

$$y_h[n]=C(3)^n$$

1.2) PARTICULAR

$$y[n]-3y[n-1]=2u[n]$$

$$y_p[n]=\beta u[n]$$

$$y_p[n]-3y_p[n-1]=2u[n]\longrightarrow\beta u[n]-3\beta u[n-1]=2u[n]\xrightarrow{n=1}\beta-3\beta=2\rightarrow\beta=-1$$

$$y[n]=C(3)^n-u[n]$$

1.3) SOLUTION

$$y[n]-3y[n-1]=x[n]$$

$$Y[0] - 3Y[-1] = x[n] \longrightarrow Y[0] - 3Y[-1] = 2\,u[0] \longrightarrow Y[0] - 3 = 2$$

$$Y[0] = C \cdot (3)^0 - u[0] = C - 1$$

$$\rightarrow C - 1 - 3 = 2 \longrightarrow C = 6$$

$$Y[n] = 6\,(3)^n - u[n] \quad , n \geq 0$$

$$Y[n] = 6\,(3)^n - 1 \quad , n \geq 0$$

2.1) ZERO INPUT

$$Y[n] - 3Y[n-1] = 0 \longrightarrow \lambda - 3 = 0 \longrightarrow \lambda = 3$$

$$Y_h[n] = C\,(3)^n$$

$$Y[0] - 3Y[-1] = 0 \rightarrow C\,(3)^0 - 3 \cdot 1 = 0 \longrightarrow C = 3$$

$$Y_{zi}[n] = 3\,(3)^n$$

2.2) ZERO STATE

$$\left.\begin{matrix} Y[n] - 3Y[n-1] = 2\,u[n] \\ Y_p[n] = \beta\,u[n] \end{matrix}\right\} \beta\,u[n] - 3\beta\,u[n-1] = 2\,u[n] \rightarrow \beta - 3\beta = 2 \rightarrow \beta = -1$$

$$Y_p[n] = -u[n]$$

$$Y_{zs}[n] = Y_h[n] + Y_p[n]\Big|_{I.C.=0} = C\,(3)^n - u[n]$$

$$Y[0] - 3\cancel{Y[-1]}^{=0} = 2\,u[0] \longrightarrow C\,(3)^0 - u[0] = 2\,u[0] \rightarrow C - 1 = 2 \rightarrow C = 3$$

$$Y_{zs}[n] = 3\,(3)^n - u[n]$$

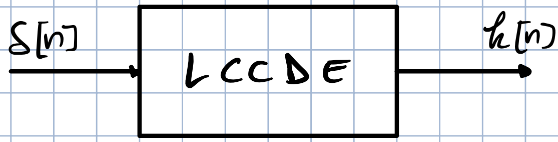
2.3) SOLUTION

$$Y[n] = Y_{zi}[n] + Y_{zs}[n] = 3(3)^n + 3\,(3)^n - u[n] = 6\,(3)^n - u[n] \quad , n \geq 0$$

$$Y[n] = 6\,(3)^n - 1 \quad , n \geq 0$$

IMPULSE RESPONSE

FOR AN LTI CAUSAL SYSTEM THE ZERO STATE SOLUTION WITH $x[n] = \delta[n]$ IS THE IMPULSE RESPONSE $h[n]$



SOLVE FOR $x[n] = \delta[n]$ AND INITIAL CONDITIONS $= 0 \ \forall n < 0$ $\left(y_p[n] = \beta\,\delta[n],\ h[n] = y_{zs}[n] \right)$

EX

$y[n] + y[n-1] - 6\,y[n-2] = x[n]$
$x[n] = \delta[n]$
$y[n] = 0 \ \forall n < 0$

1) HOMOGENEOUS

$$\lambda^2 + \lambda - 6 = 0 \longrightarrow \lambda_{1,2} = \frac{-1 \pm \sqrt{1+24}}{2} = \frac{-1 \pm 5}{2} \begin{cases} -3 \\ 2 \end{cases}$$

$$y_h[n] = C_1(-3)^n + C_2(2)^n$$

2) PARTICULAR

$$y[n] + y[n-1] - 6\,y[n-2] = x[n] \longrightarrow \beta\delta[n] + \beta\delta[n-1] - 6\beta\delta[n-2] = \delta[n] \xrightarrow{n=2} -6\beta = 0 \longrightarrow \beta = 0$$

$$y_p[n] = \beta\,\delta[n] = 0$$

3) ZERO STATE

$$y_{zs}[n] = y_h[n] + y_p[n] = C_1(-3)^n + C_2(2)^n$$

$$y[0] + \cancel{y[-1]} - \cancel{6\,y[-2]} = \delta[0] \longrightarrow C_1 + C_2 = 1$$
$$y[1] + y[0] - \cancel{6\,y[-1]} = \delta[1] \longrightarrow -3C_1 + 2C_2 + C_1 + C_2 = 0$$

$$\begin{cases} C_1 = 1 - C_2 \\ C_1 = \frac{3}{2}C_2 \end{cases} \quad \begin{cases} \frac{3}{2}C_2 = 1 - C_2 \\ C_1 = \frac{3}{2}C_2 \end{cases} \quad \begin{cases} C_2 = \frac{2}{5} \\ C_1 = \frac{3}{5} \end{cases}$$

$$y_{zs}[n] = h[n] = \frac{3}{5}(-3)^n + \frac{2}{5}(2)^n, \quad n \geq 0$$

EX

$y[n] = -3y[n-1] - 2\,y[n-2] + 2\,x[n] - x[n-2]$
$x[n] = 4^n u[n]$
$y[n] = 0 \ \forall n < 0$

1) HOMOGENEOUS

$$y[n] + 3y[n-1] + 2y[n-2] = 0 \longrightarrow \lambda^2 + 3\lambda + 2 = 0 \longrightarrow \lambda_{1,2} = \frac{-3 \pm \sqrt{9-8}}{2} = \frac{-3 \pm 1}{2} \begin{cases} -2 \\ \end{cases}$$

$$y_h[n] = C_1(-2)^n + C_2(-1)^n$$

2) PARTICULAR

$$y_p[n] = -3y_p[n-1] - 2y_p[n-2] + 2x[n] - x[n-2]$$

$$y_p[n] = \beta 4^n u[n]$$

$$\beta 4^n u[n] + 3\beta 4^{n-1} u[n-1] + 2\beta 4^{n-2} u[n-2] = 2\cdot 4^n u[n] - 4^{n-2} u[n-2]$$

$$\beta 4^2 + 3\beta 4 + 2\beta = 2\cdot 4^2 - 1$$

$$\beta(16+12+2) = 32-1$$

$$\beta = \frac{31}{30}$$

$$y_p[n] = \frac{31}{30} 4^n u[n]$$

3) SOLUTION

$$y[n] = y_h[n] + y_p[n] = C_1(-2)^n + C_2(-1)^n + \frac{31}{30}4^n u[n]$$

$$y[n] + 3y[n-1] + 2y[n-2] = 2x[n] - x[n-2]$$

$$y[0] + \cancel{3y[-1]} + \cancel{2y[-2]} = 2\cdot 4^0 u[0] - \cancel{4^{-2}u[-2]} \longrightarrow C_1(-2)^0 + C_2(-1)^0 + \frac{31}{30}4^0 u[0] = C_1 + C_2 + \frac{31}{30} = 2$$

$$y[1] + 3y[0] + \cancel{2y[-1]} = 2\cdot 4^1 u[1] - \cancel{4^{-1}u[-1]} \longrightarrow -2C_1 - C_2 + 4\cdot\frac{31}{30} + 3C_1 + 3C_2 + 3\cdot\frac{31}{30} = 8$$

$$\begin{cases} C_1 = \frac{29}{30} - C_2 \\ C_1 = \frac{23}{30} - 2C_2 \end{cases} \qquad \frac{29}{30} - C_2 = \frac{23}{30} - 2C_2 \qquad -C_2 + 2C_2 = \frac{23}{30} - \frac{29}{30} \qquad \begin{cases} C_2 = -\frac{1}{5} \\ C_1 = \frac{7}{6} \end{cases}$$

$$y[n] = \frac{7}{6}(-2)^n - \frac{1}{5}(-1)^n + \frac{31}{30}\cdot 4^n \quad , \quad n \geq 0$$